

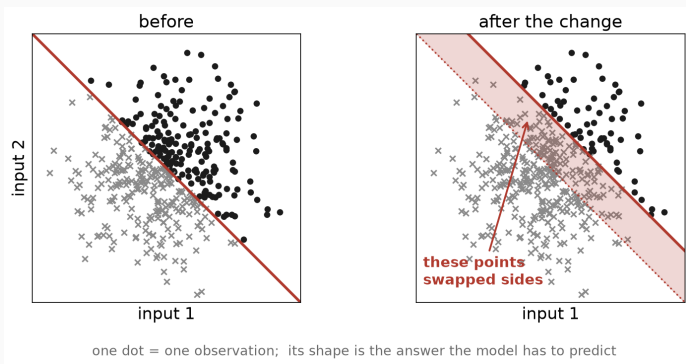
The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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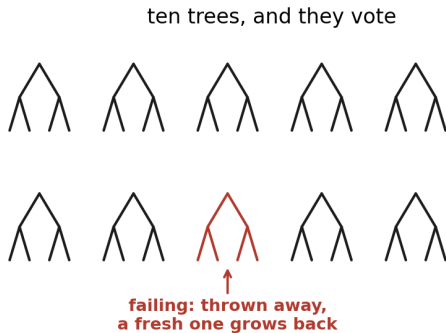
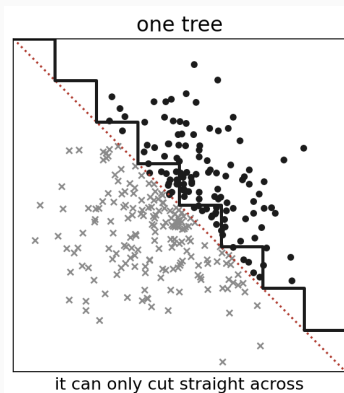
Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band**.

Defence one: a model that repairs itself

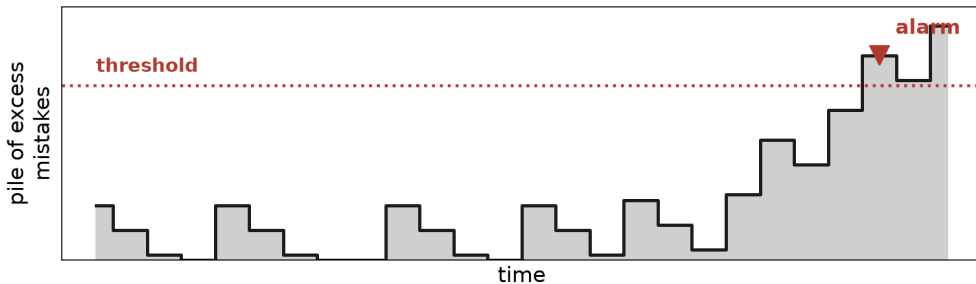


This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm

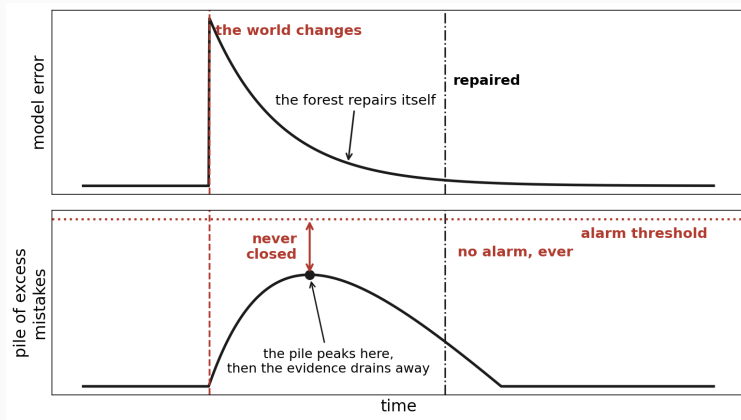
the model answers, one observation at a time (x = wrong)

x o o o x o o o o x o o o x o o x x o x x o x



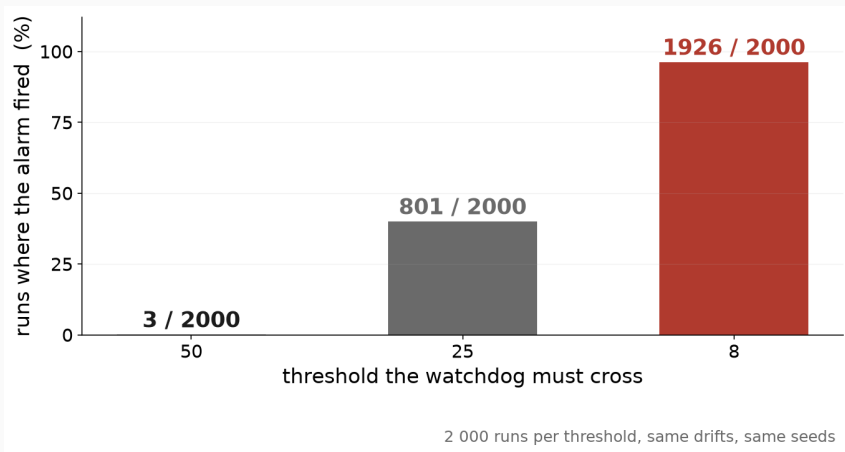
Two communities, two tools. What happens when both run at once?

Put them together, and they cancel out



The repair has a date. The alarm never gets one. (schematic)

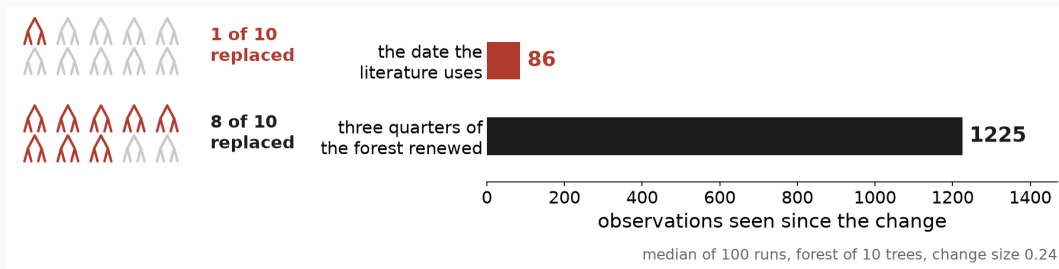
Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

Our contribution: when is the forest actually repaired?

To call the race, you have to **time the repair**.



It fires at the *first* tree. It times the start of a repair, not its end.

Measuring the repair in mistakes, not in trees

Measure	What it counts	Value
First tree replaced τ_{ARF}	the date the paper uses	86
Error back to normal τ_{rec}	the error settles back to its usual level	261
Forest 3/4 renewed τ_{swap}	8 trees out of 10 replaced	1 225
Excess mistakes $A(H)$	the extra errors the change caused	38
Watchdog's peak $S_{\text{max}}(H)$	the highest point of the watchdog's pile	33

Same change as the previous slide, median of 100 runs. First three: observations after the change. Excess mistakes: counted over the next 2 000 observations. Watchdog thresholds: 8, 25 and 50.

The paper's date says nothing about the damage.

Where we stand

at the paper's strict threshold

3 / 2 000

alarms

The blind spot is real.

the paper's stopwatch stops

up to 14×

too early

The repair looks faster
than it is.

Thank you.

a better stopwatch

?

candidates under test

The open question.